EXPERIMENT 4: IMPLEMENT K-NEAREST NEIGHBORS (KNN) AND EVALUATE MODEL PERFORMANCE

1. DATASET SOURCE

The dataset used for this experiment is the **Breast Cancer Wisconsin Dataset**.

- Source: UCI Machine Learning Repository
- Link: https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)

2. DATASET DESCRIPTION

The dataset is used for **binary classification** to predict whether a tumor is **malignant (M)** or **benign (B)**.

Basic Information

- Total Instances: **569**
- Total Features: **30 numerical features + ID + target**
- Target Variable: **diagnosis**
  - M → Malignant (Cancerous)
  - B → Benign (Non-cancerous)

Feature Categories

The features are computed from digitized images of a breast mass:

1. **Mean values** (e.g., radius_mean, texture_mean)
2. **Standard error values** (e.g., radius_se)
3. **Worst values** (e.g., radius_worst)

Example Features

- radius_mean
- texture_mean
- perimeter_mean
- area_mean
- smoothness_mean
- concavity_mean
- symmetry_mean
- fractal_dimension_mean

3. MATHEMATICAL FORMULATION OF THE ALGORITHM

KNN is a non-parametric, "lazy" learning algorithm that classifies a data point based on the majority class of its nearest neighbors.

Distance Metric

The core of the algorithm is the distance calculation between a query point and a training point. The most common metric is the **Minkowski distance**:

$$\text{dist}(x, z) = \left(\sum_{i=1}^{n} |x_i - z_i|^p\right)^{\frac{1}{p}}$$

- When $p = 1$, it is the **Manhattan Distance**.
- When $p = 2$, it is the **Euclidean Distance**.

Classification Rule

For a given test point, the algorithm:

1. Calculates distances to all training points.
2. Selects the points with the smallest distances.
3. Assigns the label that is most frequent among those neighbors (majority voting).

4. ALGORITHM LIMITATIONS

- **Computational Complexity:** Since it is a lazy learner, it does not "learn" a model during training. Instead, it performs all calculations during the prediction phase, making it slow for large datasets.
- **Memory Intensive:** It requires storing the entire training dataset in memory.
- **Sensitivity to Scale:** Features with larger ranges can dominate the distance calculation; therefore, feature scaling is mandatory.
- **Curse of Dimensionality:** In high-dimensional spaces, the "distance" between points becomes less meaningful, leading to degraded performance.

5. METHODOLOGY / WORKFLOW

Step 1: Data Preprocessing

- Removed irrelevant columns:
  - id
  - Unnamed: 32
- Converted target variable:
  - M → 1
  - B → 0

Step 2: Feature Scaling

- Applied **Standardization (Z-score normalization)**

Step 3: Train-Test Split

- Split dataset into:
  - Training set (e.g., 80%)
  - Testing set (20%)

Step 4: Model Training

- Applied **KNN classifier**
- Used Euclidean distance (default)

Step 5: Prediction

- Predicted labels on test dataset

Step 6: Evaluation

- Evaluated using:
  - Accuracy
  - Confusion Matrix
  - Precision, Recall, F1-score

6. PERFORMANCE ANALYSIS

Evaluation Metrics

1. **Accuracy**

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

2. **Precision**

$$Precision = \frac{TP}{TP + FP}$$

3. **Recall**

$$Recall = \frac{TP}{TP + FN}$$

Interpretation

- High accuracy indicates good classification performance
- High recall is important for cancer detection (minimize false negatives)
- KNN performs well due to:
    - Clean dataset
    - Numerical features
    - Balanced classes

7. HYPERPARAMETER TUNING

The most critical hyperparameter in KNN is the value of k.

- **Process:** We employ **Cross-Validation** or an **Error Rate Plot** to test various values (typically from 1 to 30).
- **Impact: * Low (e.g., k = 1):** High sensitivity to noise, leading to potential **overfitting**.
    - **High :** Smoother decision boundaries but may lead to **underfitting** by including neighbors from other classes.
- **Result:** The optimal is selected where the validation error is minimized, often an odd number to prevent ties in voting.

CONCLUSION

The implementation of the **K-Nearest Neighbors** algorithm on the **Breast Cancer dataset** demonstrates the effectiveness of distance-based classification for well-defined, low-dimensional datasets. By identifying the optimal value of through hyperparameter tuning and ensuring proper feature scaling, the model can achieve high accuracy in distinguishing between benign and malignant classes. However, the experiment also highlights that while KNN is intuitive and easy to implement, its reliance on storing all training data and calculating distances at runtime makes it less efficient for large-scale applications compared to parametric models.